Multiplex analysis-Prunus mahaleb- with Infomap

true

2023-11-23

Data

Reading the multilayer network (dataframe) with three separate files, one for each type of interaction. The data are read as square, weighted adjacency matrices to avoid problems and limitations in infomap. I'm using the `emln` setup to get a multiplex arrangement to be analizwd for multilayer modularity with `infomap` (shell binary executable).

```
# Data from raw interaction matrices.
pm.herb<- read.table(here::here("data/raw/multiplex/pmah_herb.txt"),
     na="NA", sep="\t", dec=".", header= TRUE)
pm.poll<- read.table(here::here("data/raw/multiplex/pmah_poll.txt"),
    na="NA", sep="\t", dec=".", header= TRUE)
pm.disp<- read.table(here::here("data/raw/multiplex/pmah_disp.txt"), sep="\t",
                     dec=".", header= TRUE)
pm.attr<- read.table(here::here("data/raw/multiplex/pmah_attr.txt"),
    na="NA", sep="\t", dec=".", header= TRUE)

# Interaction matrices extracted from the dataframes.
pm.herb.mat<- as.matrix(pm.herb[1:19,2:12],
        ncol = n, dimnames = list(paste0("A", 1:12), paste0("P", 1:19)))
row.names(pm.herb.mat)<- pm.herb$X[1:19]

pm.poll.mat<- as.matrix(pm.poll[1:19,2:44],
        dimnames = list(paste0("A", 1:44), paste0("P", 1:19)))
row.names(pm.poll.mat)<- pm.poll$X[1:19]

pm.disp.mat<- as.matrix(pm.disp[1:19,2:22],
        dimnames = list(paste0("A", 1:22), paste0("P", 1:19)))
row.names(pm.disp.mat)<- pm.disp$X[1:19]

# Create the layers from the matrix. These are  the intra-layer interactions,
# based on the adjacency matrices.
herbivory_layer <- pm.herb.mat
pollination_layer <- pm.poll.mat
seed_dispersal_layer <- pm.disp.mat

layer_attrib <- tibble(layer_id=1:3,
                     layer_name=c('Herbivory','Pollination',
                                  'Seed dispersal'),
                     Method=c('Observation','Collection','Observation'))

# Create the ELL tibble with interlayer links.
interlayer <- tibble(layer_from= c(rep('Herbivory',19),
```

```r
                                  rep('Pollination',19)),
                     node_from= c("pm1933","pm1931","pm1932","pm1934","pm1936",
                                  "pm1937","pm1938","pm1939","pm1940","pm1921",
                                  "pm1823","pm1824","pm1820","pm1825","pm1929",
                                  "pm8801","pm8808","pm8814","pm8865",
                                 "pm1933","pm1931","pm1932","pm1934","pm1936",
                                  "pm1937","pm1938","pm1939","pm1940","pm1921",
                                  "pm1823","pm1824","pm1820","pm1825","pm1929",
                                  "pm8801","pm8808","pm8814","pm8865"),
                     layer_to= c(rep('Pollination',19),
                                 rep('Seed dispersal',19)),
                     node_to= c("pm1933","pm1931","pm1932","pm1934","pm1936",
                                  "pm1937","pm1938","pm1939","pm1940","pm1921",
                                  "pm1823","pm1824","pm1820","pm1825","pm1929",
                                  "pm8801","pm8808","pm8814","pm8865",
                                 "pm1933","pm1931","pm1932","pm1934","pm1936",
                                  "pm1937","pm1938","pm1939","pm1940","pm1921",
                                  "pm1823","pm1824","pm1820","pm1825","pm1929",
                                  "pm8801","pm8808","pm8814","pm8865"),
                                 weight= 1) # I'm assuming here
                                            # interlayer links weights= 1.

# Create the multilayer. It has 132 nodes, counting the trees just as physical
# nodes (only once).

multilayer_bip_pru <- create_multilayer_network(list_of_layers =
                       list(herbivory_layer, pollination_layer,
                            seed_dispersal_layer),
                       bipartite = T,
                       directed = F,
                       interlayer_links = interlayer,
                       layer_attributes = layer_attrib)
```

```
## [1] "Layer #1 processing."
## [1] "Input: a bipartite matrix"
## [1] "Done."
## [1] "Layer #2 processing."
## [1] "Input: a bipartite matrix"
## [1] "Done."
## [1] "Layer #3 processing."
## [1] "Input: a bipartite matrix"
## [1] "Done."
## [1] "Creating extended link list with node IDs"
## [1] "Organizing state nodes"
```

Infomap

Exploring modules on the three monolayer networks.

These are just the original, non-square matrices for each type of interaction. I'm using this just to see the modular structure of each type of interaction analyzed with Infomap.

```r
#-----------------------------------------------------------------------------------------------
# Layer 1 - Herbivory interactions
pm.herb <- create_monolayer_network(pm.herb.mat,
                                    bipartite = T,
                                    directed = F,
                                    group_names = c('A','P'))
```

```
## [1] "Input: a bipartite matrix"
```

```r
pm.herb_object <- run_infomap_monolayer(pm.herb,
                                        infomap_executable= "Infomap",
                                        flow_model = 'undirected',
                                        silent=T, trials=20,
                                        two_level=T, seed=123)
```

```
## [1] "Creating a link list..."
## running: ./Infomap infomap.txt . --tree --seed 123 -N 20 -f undirected --silent --two-level
## [1] "Removing auxilary files..."
```

```r
colnames(pm.herb_object$edge_list)<- c("from","to","w")
# my_plot_modular_matrix(pm.herb_object)

#-----------------------------------------------------------------------------------------------
# Layer 2 - Pollination interactions
pm.poll <- create_monolayer_network(pm.poll.mat,
                                    bipartite = T,
                                    directed = F,
                                    group_names = c('A','P'))
```

```
## [1] "Input: a bipartite matrix"
```

```r
pm.poll_object <- run_infomap_monolayer(pm.poll,
                                        infomap_executable= "Infomap",
                                        flow_model = 'undirected',
                                        silent=T, trials=20,
                                        two_level=T, seed=123)
```

```
## [1] "Creating a link list..."
## running: ./Infomap infomap.txt . --tree --seed 123 -N 20 -f undirected --silent --two-level
## [1] "Removing auxilary files..."
```

```r
#colnames(pm.poll_object$edge_list)<- c("from","to","w")
# my_plot_modular_matrix(pm.poll_object)
#-----------------------------------------------------------------------------------------------
# Layer 3 - Seed dispersal interactions
pm.disp <- create_monolayer_network(pm.disp.mat,
                                    bipartite = T,
                                    directed = F,
                                    group_names = c('A','P'))
```

```
## [1] "Input: a bipartite matrix"
```

```r
pm.disp_object <- run_infomap_monolayer(pm.disp,
                                        infomap_executable= "Infomap",
                                        flow_model = 'undirected',
                                        silent=T, trials=20,
                                        two_level=T, seed=123)
```

```
## [1] "Creating a link list..."
## running: ./Infomap infomap.txt . --tree --seed 123 -N 20 -f undirected --silent --two-level
## [1] "Removing auxilary files..."
```

```
# my_plot_modular_matrix(pm.disp_object)
```

Multilayer network

Now, we will create the multilayer network. Although effectively it is a bipartite network, we will treat it as an unipartite network to be able to account for link directions. The description of function `create_multilayer_network` is in the package emln. Plotting is done using two dedicated functions in infomapecology: `plot_multilayer_modules` and `plot_multilayer_alluvial`.

```
# Tree atributes dataset.
# Attributes file.
pm.attr<- read.table(here("./data/raw/prunus_data/pmah_attr.txt"),
                     sep="\t", dec=".", header= TRUE)

require(forcats)
#pm.attr$pollen <- as.numeric(fct_relevel(data$ISLAND, "GC","TF","Go"))
pm.attr$pollen <- as.factor(fct_relevel(pm.attr$pollen, "n","+","++"))

#
# Create a multilayer object
#
layer_attrib <- tibble(layer_id=1:3, layer_name=c('Herbivory','Pollination','Dispersal'))
pmah_interlayer <- read.table(here("./data/raw/multiplex/pmah_interlayer.csv"),
                  header=TRUE, sep=",", dec = ".", quote = "", comment.char = "#")
pmah_layer_attr<- layer_attrib

multilayer_prunus <- create_multilayer_network(list_of_layers = list(pm.herb.mat,
                                                                     pm.poll.mat,
                                                                     pm.disp.mat),
                                               layer_attributes = pmah_layer_attr,
                                               interlayer_links = pmah_interlayer,
                                               bipartite = T,
                                               directed = F,
                                               physical_node_attributes = pm.attr)
```

```
## [1] "Layer #1 processing."
## [1] "Input: a bipartite matrix"
## [1] "Done."
## [1] "Layer #2 processing."
## [1] "Input: a bipartite matrix"
## [1] "Done."
## [1] "Layer #3 processing."
## [1] "Input: a bipartite matrix"
## [1] "Done."
## [1] "Organizing state nodes"
## [1] "Creating extended link list with node IDs"
## [1] "Organizing state nodes"
```

Multilayer community detection

```
source(here::here("RCode/functions/my_run_infomap_multilayer.R"))
system("./Infomap ./out/multimap_prunus_infomap.edges ./out/ --clu --seed 12345 --num-trials 50 -v --mul

multilayer_bip_pru_modularity <- my_run_infomap_multilayer(M = multilayer_bip_pru,
                            relax = TRUE, silent = FALSE,
                            flow_model = "undirected",
                            trials = 50, seed = 497294,
                            twoLevel=T,
                            preclusterMultiplex=F,
                            addMissingPhysicalNodes=F,
                            multilayer_relax_rate = 0.01,
                            multilayer_relax_limit_up = 0.5,
                            multilayer_relax_limit_down = 0,
                            multilayerJSRelaxRate=NA,
                            temporal_network = FALSE)
```

```
## Warning: `as.tbl()` was deprecated in dplyr 1.0.0.
## i Please use `tibble::as_tibble()` instead.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## [1] "Using global relax to determine flow between layers."
## [1] "./Infomap infomap_multilayer.txt . --tree -2 -N 50 --seed 497294 -f undirected TRUE FALSE FALSE
## [1] "Removing auxilary files..."
## [1] "Partitioned into 7 modules."
```

```
# List of module composition
multilayer_bip_pru_modularity$m
```

```
## [1] 7
```

```
module.comp<- print(multilayer_bip_pru_modularity$modules %>%
    dplyr::select(node_id, layer_id, node_name, module), n=1000)
```

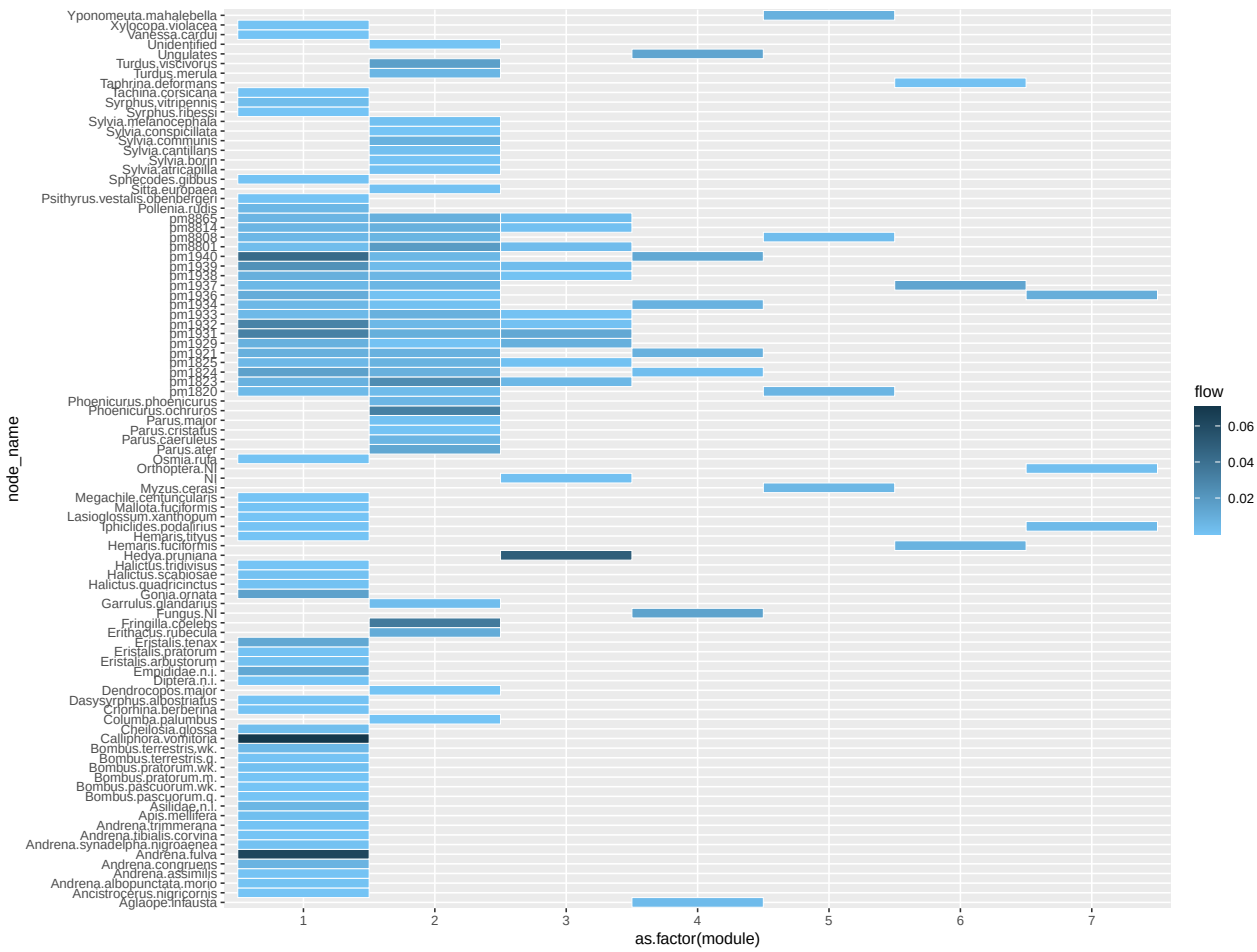
```
## # A tibble: 132 x 4
##     node_id layer_id node_name                      module
##       <int>    <int> <chr>                           <int>
##   1       1        1 Aglaope.infausta                    4
##   2       2        2 Ancistrocerus.nigricornis           1
##   3       3        2 Andrena.albopunctata.morio          1
##   4       4        2 Andrena.assimilis                   1
##   5       5        2 Andrena.congruens                   1
##   6       6        2 Andrena.fulva                       1
##   7       7        2 Andrena.synadelpha.nigroaenea       1
##   8       8        2 Andrena.tibialis.corvina            1
##   9       9        2 Andrena.trimmerana                  1
##  10      10        2 Apis.mellifera                      1
##  11      11        2 Asilidae.n.i.                       1
##  12      12        2 Bombus.pascuorum.q.                 1
##  13      13        2 Bombus.pascuorum.wk.                1
##  14      14        2 Bombus.pratorum.m.                  1
##  15      15        2 Bombus.pratorum.wk.                 1
##  16      16        2 Bombus.terrestris.q.                1
##  17      17        2 Bombus.terrestris.wk.               1
```

```
## 18      18        2 Calliphora.vomitoria              1
## 19      19        2 Cheilosia.glossa                  1
## 20      20        3 Columba.palumbus                  2
## 21      21        2 Criorhina.berberina               1
## 22      22        2 Dasysyrphus.albostriatus          1
## 23      23        3 Dendrocopos.major                 2
## 24      24        2 Diptera.n.i.                      1
## 25      25        2 Empididae.n.i.                    1
## 26      26        2 Eristalis.arbustorum              1
## 27      27        2 Eristalis.pratorum                1
## 28      28        2 Eristalis.tenax                   1
## 29      29        3 Erithacus.rubecula                2
## 30      30        3 Fringilla.coelebs                 2
## 31      31        1 Fungus.NI                         4
## 32      32        3 Garrulus.glandarius               2
## 33      33        2 Gonia.ornata                      1
## 34      34        2 Halictus.quadricinctus            1
## 35      35        2 Halictus.scabiosae                1
## 36      36        2 Halictus.tridivisus               1
## 37      37        1 Hedya.pruniana                    3
## 38      38        1 Hemaris.fuciformis                6
## 39      39        2 Hemaris.tityus                    1
## 40      40        1 Iphiclides.podalirius             7
## 41      40        2 Iphiclides.podalirius             1
## 42      41        2 Lasioglossum.xanthopum            1
## 43      42        2 Mallota.fuciformis                1
## 44      43        2 Megachile.centuncularis           1
## 45      44        1 Myzus.cerasi                      5
## 46      45        1 NI                                3
## 47      46        1 Orthoptera.NI                     7
## 48      47        2 Osmia.rufa                        1
## 49      48        3 Parus.ater                        2
## 50      49        3 Parus.caeruleus                   2
## 51      50        3 Parus.cristatus                   2
## 52      51        3 Parus.major                       2
## 53      52        3 Phoenicurus.ochruros              2
## 54      53        3 Phoenicurus.phoenicurus           2
## 55      54        1 pm1820                            5
## 56      54        2 pm1820                            1
## 57      54        3 pm1820                            2
## 58      55        1 pm1823                            3
## 59      55        2 pm1823                            1
## 60      55        3 pm1823                            2
## 61      56        1 pm1824                            4
## 62      56        2 pm1824                            1
## 63      56        3 pm1824                            2
## 64      57        1 pm1825                            3
## 65      57        2 pm1825                            1
## 66      57        3 pm1825                            2
## 67      58        1 pm1921                            4
## 68      58        2 pm1921                            1
## 69      58        3 pm1921                            2
## 70      59        1 pm1929                            3
## 71      59        2 pm1929                            1
```

```
## 72      59        3 pm1929                              2
## 73      60        1 pm1931                              3
## 74      60        2 pm1931                              1
## 75      60        3 pm1931                              2
## 76      61        1 pm1932                              3
## 77      61        2 pm1932                              1
## 78      61        3 pm1932                              2
## 79      62        1 pm1933                              3
## 80      62        2 pm1933                              1
## 81      62        3 pm1933                              2
## 82      63        1 pm1934                              4
## 83      63        2 pm1934                              1
## 84      63        3 pm1934                              2
## 85      64        1 pm1936                              7
## 86      64        2 pm1936                              1
## 87      64        3 pm1936                              2
## 88      65        1 pm1937                              6
## 89      65        2 pm1937                              1
## 90      65        3 pm1937                              2
## 91      66        1 pm1938                              3
## 92      66        2 pm1938                              1
## 93      66        3 pm1938                              2
## 94      67        1 pm1939                              3
## 95      67        2 pm1939                              1
## 96      67        3 pm1939                              2
## 97      68        1 pm1940                              4
## 98      68        2 pm1940                              1
## 99      68        3 pm1940                              2
## 100     69        1 pm8801                              3
## 101     69        2 pm8801                              1
## 102     69        3 pm8801                              2
## 103     70        1 pm8808                              5
## 104     70        2 pm8808                              1
## 105     70        3 pm8808                              2
## 106     71        1 pm8814                              3
## 107     71        2 pm8814                              1
## 108     71        3 pm8814                              2
## 109     72        1 pm8865                              3
## 110     72        2 pm8865                              1
## 111     72        3 pm8865                              2
## 112     73        2 Pollenia.rudis                      1
## 113     74        2 Psithyrus.vestalis.obenbergeri      1
## 114     75        3 Sitta.europaea                      2
## 115     76        2 Sphecodes.gibbus                    1
## 116     77        3 Sylvia.atricapilla                  2
## 117     78        3 Sylvia.borin                        2
## 118     79        3 Sylvia.cantillans                   2
## 119     80        3 Sylvia.communis                     2
## 120     81        3 Sylvia.conspicillata                2
## 121     82        3 Sylvia.melanocephala                2
## 122     83        2 Syrphus.ribessi                     1
## 123     84        2 Syrphus.vitripennis                 1
## 124     85        2 Tachina.corsicana                   1
## 125     86        1 Taphrina.deformans                  6
```

```
## 126        87          3 Turdus.merula                      2
## 127        88          3 Turdus.viscivorus                  2
## 128        89          1 Ungulates                          4
## 129        90          3 Unidentified                       2
## 130        91          2 Vanessa.cardui                     1
## 131        92          2 Xylocopa.violacea                  1
## 132        93          1 Yponomeuta.mahalebella             5
```

```
module.comp<- as.data.frame(module.comp)
#-----------------------------------------------------------------------
ggplot(multilayer_bip_pru_modularity$modules, aes(x= as.factor(module),
                                                  y= node_name))+
                                                  geom_tile(aes(fill = flow),
                                colour = "white") +
                                scale_fill_gradient(low = "#75C4F5",
                                high = "#14374B")
```



Module assignement 3D plot

```
# 3d heatmap
#
data<- multilayer_bip_pru_modularity$modules
str(data)
```

```
## tibble [132 x 5] (S3: tbl_df/tbl/data.frame)
##  $ node_id  : int [1:132] 1 2 3 4 5 6 7 8 9 10 ...
##  $ module   : int [1:132] 4 1 1 1 1 1 1 1 1 1 ...
##  $ layer_id : int [1:132] 1 2 2 2 2 2 2 2 2 2 ...
##  $ flow     : num [1:132] 0.004008 0.000118 0.000792 0.000424 0.007726 ...
##  $ node_name: chr [1:132] "Aglaope.infausta" "Ancistrocerus.nigricornis" "Andrena.albopunctata.morio
```

```r
# Unstacking using pivot_wider()
df_wide <- pivot_wider(data, id_cols = node_name, names_from = module, values_from = flow)
str(df_wide)
```

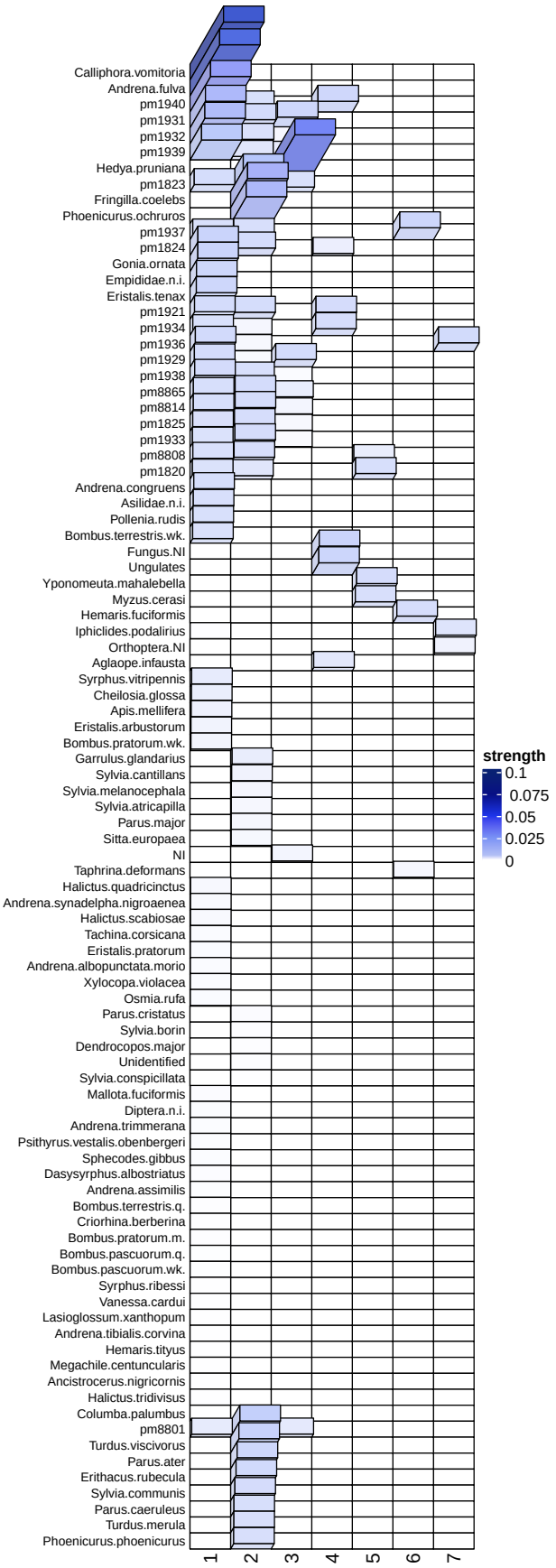
```
## tibble [93 x 8] (S3: tbl_df/tbl/data.frame)
##  $ node_name: chr [1:93] "Aglaope.infausta" "Ancistrocerus.nigricornis" "Andrena.albopunctata.morio"
##  $ 4        : num [1:93] 0.00401 NA NA NA NA ...
##  $ 1        : num [1:93] NA 0.000118 0.000792 0.000424 0.007726 ...
##  $ 2        : num [1:93] NA NA NA NA NA NA NA NA NA NA ...
##  $ 3        : num [1:93] NA NA NA NA NA NA NA NA NA NA ...
##  $ 6        : num [1:93] NA NA NA NA NA NA NA NA NA NA ...
##  $ 7        : num [1:93] NA NA NA NA NA NA NA NA NA NA ...
##  $ 5        : num [1:93] NA NA NA NA NA NA NA NA NA NA ...
```

```r
head(df_wide)
```

```
## # A tibble: 6 x 8
##   node_name                      `4`       `1`   `2`   `3`   `6`   `7`   `5`
##   <chr>                        <dbl>     <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Aglaope.infausta            0.00401 NA           NA    NA    NA    NA    NA
## 2 Ancistrocerus.nigricornis  NA        0.000118    NA    NA    NA    NA    NA
## 3 Andrena.albopunctata.morio NA        0.000792    NA    NA    NA    NA    NA
## 4 Andrena.assimilis          NA        0.000424    NA    NA    NA    NA    NA
## 5 Andrena.congruens          NA        0.00773     NA    NA    NA    NA    NA
## 6 Andrena.fulva              NA        0.0620      NA    NA    NA    NA    NA
```

```r
df_wide.mat<- df_wide %>% select(3,4,5,2,8,6,7)
df_wide.mat<- as.data.frame(df_wide.mat)
df_wide.mat<- as.matrix(df_wide.mat)
rownames(df_wide.mat)<- df_wide$node_name
# Replace NA values with 0
df_wide.mat[is.na(df_wide.mat)] <- 0

# Plot
col_fun = circlize::colorRamp2(c(0, 0.005, 0.025, 0.05, 0.075, 0.1),
                               c("#FFFFFF", "#B3C0F5", "#7A93F5",
                                 "#1E36EB", "#070F85","#082173"))
ht_opt$TITLE_PADDING = unit(c(10, 2), "mm")
Heatmap3D(log(df_wide.mat+1), name = "strength", col = col_fun,
          cluster_columns = FALSE, show_row_dend = FALSE,
          show_column_names = TRUE,
          heatmap_legend_param = list(at = c(0, 0.025, 0.05, 0.075, 0.1),
                                      labels = c("0", "0.025", "0.05", " 0.075", "0.1")),
          column_title = '       ',
          row_names_side = "left",
          row_names_gp = gpar(fontsize = 8),
          # new arguments for Heatmap3D()
          bar_rel_width = 1, bar_rel_height = 1, bar_max_length = unit(1.5, "cm")
          )
```


Calliphora.vomitoria
Andrena.fulva
pm1940
pm1931
pm1932
pm1939
Hedya.pruniana
pm1823
Fringilla.coelebs
Phoenicurus.ochruros
pm1937
pm1824
Gonia.ornata
Empididae.n.i.
Eristalis.tenax
pm1921
pm1934
pm1936
pm1929
pm1938
pm8865
pm8814
pm1825
pm1933
pm8808
pm1820
Andrena.congruens
Asilidae.n.i.
Pollenia.rudis
Bombus.terrestris.wk.
Fungus.NI
Ungulates
Yponomeuta.mahalebella
Myzus.cerasi
Hemaris.fuciformis
Iphiclides.podalirius
Orthoptera.NI
Aglaope.infausta
Syrphus.vitripennis
Cheilosia.glossa
Apis.mellifera
Eristalis.arbustorum
Bombus.pratorum.wk.
Garrulus.glandarius
Sylvia.cantillans
Sylvia.melanocephala
Sylvia.atricapilla
Parus.major
Sitta.europaea
NI
Taphrina.deformans
Halictus.quadricinctus
Andrena.synadelpha.nigroaenea
Halictus.scabiosae
Tachina.corsicana
Eristalis.pratorum
Andrena.albopunctata.morio
Xylocopa.violacea
Osmia.rufa
Parus.cristatus
Sylvia.borin
Dendrocopos.major
Unidentified
Sylvia.conspicillata
Mallota.fuciformis
Diptera.n.i.
Andrena.trimmerana
Psithyrus.vestalis.obenbergeri
Sphecodes.gibbus
Dasysyrphus.albostriatus
Andrena.assimilis
Bombus.terrestris.q.
Criorhina.berberina
Bombus.pratorum.m.
Bombus.pascuorum.q.
Bombus.pascuorum.wk.
Syrphus.ribessi
Vanessa.cardui
Lasioglossum.xanthopum
Andrena.tibialis.corvina
Hemaris.tityus
Megachile.centuncularis
Ancistrocerus.nigricornis
Halictus.tridivisus
Columba.palumbus
pm8801
Turdus.viscivorus
Parus.ater
Erithacus.rubecula
Sylvia.communis
Parus.caeruleus
Turdus.merula
Phoenicurus.phoenicurus
1
2
3
4
5
6
7
strength
0.1
0.075
0.05
0.025
0

Thus modular arrangement can be compared with the one obtained with `bipartite` on the flattened multilayer (in `emln` analysis).

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R packages used

Package	Version	Citation
attempt	0.3.1	Fay (2020)
base	4.4.3	R Core Team (2025)
bipartite	2.21	(**biparti?**)....
ComplexHeatmap	2.22.0	(**Complex?**)....
cowplot	1.1.3	Wilke (2024)
devtools	2.4.5	Wickham et al. (2022)
DT	0.33	Xie, Cheng, and Tan (2024)
emln	1.0.2	Frydman, Pilosof, and Freilikhman (2025)
ggalluvial	0.12.5	(**ggalluv?**)....
GGally	2.2.1	Schloerke et al. (2024)
ggbipart	0.1.2	Jordano (2025)
ggcorrplot	0.1.4.1	Kassambara (2023)
ggraph	2.2.1	Pedersen (2024)
hablar	0.3.2	Sjoberg (2023)
here	1.0.1	Müller (2020)
igraph	2.1.4	(**igraph2?**)....
infomapecology	2.0	(**infomap?**)....
intergraph	2.0.4	Bojanowski (2023)
knitr	1.50	(**knitr20?**)....
lattice	0.22.7	Sarkar (2008)
lme4	1.1.37	Bates et al. (2015)
Matrix	1.7.3	Bates, Maechler, and Jagan (2025)
mgcv	1.9.3	(**mgcv200?**)....
multinet	4.2.1	Magnani, Rossi, and Vega (2021)
MuMIn	1.48.11	Bartoń (2025)
muxViz	3.1	De Domenico (2025)
network	1.19.0	Butts (2008)....
nlme	3.1.168	(**nlme200?**)....
performance	0.13.0	Lüdecke et al. (2021)
permute	0.9.7	Simpson (2022)
RColorBrewer	1.1.3	(**RColorB?**)....
Rcpp	1.0.14	(**Rcpp201?**)....
rgl	1.3.18	Murdoch and Adler (2025)
rsq	2.7	Zhang (2024)
sna	2.8	Butts (2024)
statnet.common	4.11.0	(**statnet?**)....
tidyverse	2.0.0	Wickham et al. (2019)
usethis	3.1.0	Wickham et al. (2024)
vegan	2.6.10	Oksanen et al. (2025)
viridis	0.6.5	Garnier et al. (2024)
viridisLite	0.4.2	(**viridisLite?**)

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Session info

```r
sessionInfo()
```

```
## R version 4.4.3 (2025-02-28)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sequoia 15.4
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Madrid
## tzcode source: internal
##
## attached base packages:
## [1] grid      stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] ComplexHeatmap_2.22.0 infomapecology_2.0    emln_1.0.2
##  [4] hablar_0.3.2          DT_0.33               rlang_1.1.5
##  [7] magrittr_2.0.3        ggalluvial_0.12.5     cowplot_1.1.3
## [10] attempt_0.3.1         ggcorrplot_0.1.4.1    ggbipart_0.1.2
## [13] ggraph_2.2.1          intergraph_2.0-4      GGally_2.2.1
## [16] lme4_1.1-37           Matrix_1.7-3          viridis_0.6.5
## [19] viridisLite_0.4.2     rgl_1.3.18            rsq_2.7
## [22] muxViz_3.1            performance_0.13.0    mgcv_1.9-3
## [25] nlme_3.1-168          MuMIn_1.48.11         multinet_4.2.1
## [28] RColorBrewer_1.1-3    Rcpp_1.0.14           igraph_2.1.4
## [31] bipartite_2.21        vegan_2.6-10          lattice_0.22-7
## [34] permute_0.9-7         sna_2.8               network_1.19.0
## [37] statnet.common_4.11.0 knitr_1.50            devtools_2.4.5
## [40] usethis_3.1.0         lubridate_1.9.4       forcats_1.0.0
## [43] stringr_1.5.1         dplyr_1.1.4           purrr_1.0.4
## [46] readr_2.1.5           tidyr_1.3.1           tibble_3.2.1
## [49] ggplot2_3.5.1         tidyverse_2.0.0       here_1.0.1
##
## loaded via a namespace (and not attached):
##   [1] splines_4.4.3       later_1.4.1         fields_16.3.1
##   [4] polyclip_1.10-7     lifecycle_1.0.4     Rdpack_2.6.3
##   [7] doParallel_1.0.17   rprojroot_2.0.4     vroom_1.6.5
##  [10] MASS_7.3-65         insight_1.1.0       rmarkdown_2.29
##  [13] yaml_2.3.10         remotes_2.5.0       httpuv_1.6.15
##  [16] grateful_0.2.11     robslopes_1.1.3     spam_2.11-1
##  [19] sessioninfo_1.2.3   pkgbuild_1.4.7      minqa_1.2.8
##  [22] maps_3.4.2.1        pkgload_1.4.0       BiocGenerics_0.52.0
##  [25] tweenr_2.0.3        circlize_0.4.16     IRanges_2.40.1
##  [28] S4Vectors_0.44.0    ggrepel_0.9.6       codetools_0.2-20
##  [31] ggforce_0.4.2       tidyselect_1.2.1    shape_1.4.6.1
##  [34] farver_2.1.2        matrixStats_1.5.0   stats4_4.4.3
```

```
## [37] base64enc_0.1-3     jsonlite_2.0.0      GetoptLong_1.0.5
## [40] ellipsis_0.3.2      tidygraph_1.3.1     iterators_1.0.14
## [43] foreach_1.5.2       tools_4.4.3         glue_1.8.0
## [46] gridExtra_2.3       Rttf2pt1_1.3.12     xfun_0.52
## [49] withr_3.0.2         fastmap_1.2.0       boot_1.3-31
## [52] digest_0.6.37       timechange_0.3.0    R6_2.6.1
## [55] mime_0.13           colorspace_2.1-1    Cairo_1.6-2
## [58] utf8_1.2.4          generics_0.1.3      graphlayouts_1.2.2
## [61] htmlwidgets_1.6.4   ggstats_0.9.0       pkgconfig_2.0.3
## [64] gtable_0.3.6        htmltools_0.5.8.1   profvis_0.4.0
## [67] dotCall64_1.2       clue_0.3-66         scales_1.3.0
## [70] png_0.1-8           reformulas_0.4.0    deming_1.4-1
## [73] rstudioapi_0.17.1   tzdb_0.5.0          rjson_0.2.23
## [76] coda_0.19-4.1       nloptr_2.2.1        cachem_1.1.0
## [79] GlobalOptions_0.1.2 parallel_4.4.3      miniUI_0.1.1.1
## [82] extrafont_0.19      pillar_1.10.2       vctrs_0.6.5
## [85] urlchecker_1.0.1    promises_1.3.2      xtable_1.8-4
## [88] Deriv_4.1.6         cluster_2.1.8.1     extrafontdb_1.0
## [91] evaluate_1.0.3      magick_2.8.6        tinytex_0.56
## [94] cli_3.6.4           compiler_4.4.3      crayon_1.5.3
## [97] labeling_0.4.3      plyr_1.8.9          fs_1.6.5
## [100] stringi_1.8.7      mcr_1.3.3.1         munsell_0.5.1
## [103] hms_1.1.3          bit64_4.6.0-1       shiny_1.10.0
## [106] rbibutils_2.3      memoise_2.0.1       bit_4.6.0
```

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